Susquehanna International Group Data Engineer interview questions & guide 2026

Every question Susquehanna International Group interviewers actually ask, the frameworks that win the room, and the language hiring managers respond to.

What is a Data Engineer at Susquehanna International Group?

At **Susquehanna International Group (SIG)**, data is the fundamental driver of every trading decision, quantitative model, and risk assessment. As a **Data Engineer**, you do not simply move data from one repository to another; you design, build, and optimize the high-performance pipelines that process massive volumes of market and proprietary data. Your work directly impacts the speed and accuracy with which **SIG** can identify market opportunities and execute trades in highly competitive global markets.

In this role, you will collaborate closely with quantitative researchers, traders, and software developers to translate complex financial data into clean, accessible, and highly optimized datasets. The systems you build must handle extreme scale and maintain near-zero latency, requiring a deep understanding of hardware limitations, memory management, and algorithmic efficiency. Whether you are optimizing data ingestion frameworks or refining historical databases for backtesting, your contributions are critical to maintaining **SIG**'s competitive edge.

To succeed as a **Data Engineer** at **SIG**, you must possess a passion for solving intricate performance bottlenecks and a desire to understand the inner workings of

your tools. This is an environment that rewards technical curiosity, analytical rigor, and a proactive approach to engineering challenges. If you thrive on building robust systems where every millisecond and byte of memory matters, this position offers an incredibly rewarding intellectual and professional playground.

Common Interview Questions

To help you prepare effectively, we have compiled representative questions based on real candidate experiences at **Susquehanna International Group**. These questions highlight the core competencies evaluated during the interview process, specifically focusing on advanced Python programming, deep library knowledge, and data engineering fundamentals.

Python & Core Programming

This category tests your mastery of Python beyond basic syntax, focusing on language internals, memory management, and advanced programming paradigms.

- Explain how Python decorators work under the hood and write a custom decorator to log the execution time of a function.
- What is the difference between a generator and a list comprehension in terms of memory utilization and execution?
- How does Python's Global Interpreter Lock (GIL) impact multi-threaded data processing, and how can you bypass its limitations?
- Describe the difference between deep copy and shallow copy in Python, and explain when you would use each.
- How does Python manage memory, and what role does the garbage collector play in reference counting?

Data Manipulation & Pandas Internals

These questions evaluate your ability to manipulate large datasets efficiently and your understanding of how data structures are managed in memory.

- Explain how **Pandas** manages data under the hood. What is the **BlockManager**, and how does it impact memory usage?
- How would you optimize a slow **Pandas** operation that performs row-by-row calculations on a dataframe containing millions of rows?
- Describe the difference between `loc`, `iloc`, and basic indexing in **Pandas** in terms of performance and behavior.
- How does **Pandas** handle missing data (NaNs) across different data types, and what are the performance implications of casting?
- Explain the concept of vectorization in **Pandas** and why it is significantly faster than standard Python loops.

Data Engineering Practices & Architecture

These questions assess your ability to design robust, scalable, and maintainable data pipelines and storage systems.

- How would you design an ETL pipeline to ingest and process high-frequency market data with minimal latency?
- What factors do you consider when choosing between row-oriented (e.g., Avro) and column-oriented (e.g., Parquet) storage formats?
- Describe your approach to implementing data quality checks and monitoring in a production data pipeline.
- How do you handle schema evolution when updating data pipelines that feed downstream quantitative models?
- Explain how you would optimize a slow-running SQL query that joins several large transaction tables.

The questions most likely to come up

Sorted by relevance to this company

#	Question	Topic	Difficulty	Asked
01	Calculate Monthly Sales Growth by Product Category	SQL & Data Manipulation	Medium	Very common
02	Multi-Level Aggregations in SQL	SQL & Data Manipulation	Medium	Very common
03	Choosing INNER vs LEFT JOIN	SQL & Data Manipulation	Medium	Very common
04	Troubleshoot Nightly Orders Pipeline Failure	Pipelines	Easy	Common
05	Understanding Database Indexing Purpose	SQL & Data Manipulation	Medium	Very common
06	Discuss Core SQL Query Experience	SQL & Data Manipulation	Easy	Very common
07	Data Quality in ETL Pipelines	Pipelines	Easy	Very common
08	Prioritizing Across Competing Client Projects	Behavioral & Leadership	Easy	Very common
09	Handling Missing Data in Pipelines	Pipelines	Medium	Very common
10	Prioritizing Conflicting High-Stakes Work	Behavioral & Leadership	Easy	Very common
11	Choosing Batch vs Real Time	Pipelines	Hard	Very common
12	Leading Through an Ambiguous Project Crisis	Behavioral & Leadership	Medium	Very common
13	Cloud AI Deployment Pipeline Experience	Pipelines	Medium	Very common
14	Diagnose an Integration Pipeline Issue	Pipelines	Medium	Very common
15	Influencing a Cross-Functional Decision	Behavioral & Leadership	Medium	Very common
16	Real-Time Dashboard Data Pipeline	Pipelines	Medium	Very common
17	Explaining Technical Complexity to Business Partners	Behavioral & Leadership	Easy	Very common
18	Resolving Conflict Within Your Team	Behavioral & Leadership	Easy	Very common
19	Resolving Conflict on a Client Account	Behavioral & Leadership	Medium	Very common
20	Integrating Multiple Data Sources	Pipelines	Easy	Very common

Getting Ready for Your Interviews

Preparing for an interview at **Susquehanna International Group** requires a shift in mindset from general software development to high-performance data manipulation. You should focus on demonstrating not just that you can write code that works, but that you understand exactly how that code executes at the system level.

Technical Rigor – You must demonstrate a granular understanding of your primary tools. Do not just memorize APIs; know how **Python** and **Pandas** manage memory, how data structures are laid out, and how to write highly optimized code.

Problem-Solving & Efficiency – Interviewers will evaluate how you approach complex, ambiguous data challenges. They want to see you break down a problem systematically, identify performance bottlenecks early, and propose highly efficient solutions.

Collaboration & Communication – Because **Data Engineers** work closely with trading desks and quantitative researchers, you must be able to explain complex technical trade-offs clearly. Show that you can translate business requirements into technical specifications seamlessly.

Tip

SIG values deep technical understanding over superficial framework knowledge. Be prepared to explain how tools work under the hood, not just how to use them.

Interview Process Overview

The interview process for a **Data Engineer** at **Susquehanna International Group** is structured to thoroughly evaluate both your technical depth and your alignment with the team's collaborative culture. The process is rigorous but highly structured, typically progressing from initial technical assessments to intensive, multi-round technical evaluations.

The journey begins with an initial phone screen or technical assessment focusing on core programming concepts, specifically in **Python**. If you pass this stage, you will move on to dedicated technical interviews that dive deep into data manipulation with **Pandas** and advanced **Python** paradigms. These rounds are designed to test your hands-on coding ability and your understanding of data structures.

The final stage is an intensive loop—often conducted on-site at offices like **Dublin, Ireland** or **Bala Cynwyd, Pennsylvania**. This final stage consists of multiple consecutive rounds with engineering team members, technical leaders, and sometimes security or infrastructure partners. These conversations focus on system design, pipeline architecture, and your ability to solve real-world data scaling challenges in real time.